

**Building Resilient IT Infrastructure Against Cyberattacks in  
Nigeria: A Cross-Sector Analysis of Banks, Government  
Agencies, and Critical Infrastructure**

**BABATUNDE OJO**

**2025/A/MIT/0126**

**301800742**

**MIT 8212**

**Miva Open University**

**July 3rd 2026**

## Table of Contents

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>ABSTRACT .....</b>                                                 | <b>4</b>  |
| <b>1. Introduction.....</b>                                           | <b>5</b>  |
| <b>1.1 Background .....</b>                                           | <b>5</b>  |
| <b>1.2 The ByteToBreach Incident: A Cross-Sector Case Study .....</b> | <b>6</b>  |
| <b>1.3 Timeline of Events .....</b>                                   | <b>7</b>  |
| <b>1.4 Problem Statement.....</b>                                     | <b>9</b>  |
| <b>1.5 Research Question .....</b>                                    | <b>9</b>  |
| <b>1.6 Objectives.....</b>                                            | <b>9</b>  |
| <b>1.7 Scope, Significance, and Structure .....</b>                   | <b>10</b> |
| <b>2. Literature and Industry Review .....</b>                        | <b>11</b> |
| <b>2.1 Global Trends in IT Infrastructure Resilience .....</b>        | <b>11</b> |
| <b>2.2 The Nigerian Regulatory and Institutional Landscape .....</b>  | <b>12</b> |
| <b>2.3 Sector-Specific Literature.....</b>                            | <b>14</b> |
| <b>2.3.1 Banking Sector .....</b>                                     | <b>14</b> |
| <b>2.3.2 Government Sector .....</b>                                  | <b>14</b> |
| <b>2.4 Theoretical Frameworks .....</b>                               | <b>15</b> |
| <b>3. Investigative Approach .....</b>                                | <b>17</b> |
| <b>3.1 Research Design.....</b>                                       | <b>17</b> |
| <b>3.2 Data Sources.....</b>                                          | <b>17</b> |
| <b>3.3 Analytical Tools .....</b>                                     | <b>18</b> |
| <b>3.4 Ethical Considerations and Limitations .....</b>               | <b>18</b> |
| <b>4. Findings and Analysis.....</b>                                  | <b>19</b> |
| <b>4.1 NIST CSF Gap Analysis: Cross-Case Findings.....</b>            | <b>19</b> |
| <b>4.2 Root Cause Analysis.....</b>                                   | <b>21</b> |
| <b>4.2.1 Technical Root Causes .....</b>                              | <b>21</b> |
| <b>4.2.2 Organisational Root Causes .....</b>                         | <b>22</b> |
| <b>4.2.3 Regulatory Root Causes .....</b>                             | <b>22</b> |
| <b>4.2.4 Structural Root Causes .....</b>                             | <b>23</b> |
| <b>4.3 Sector SWOT Analysis .....</b>                                 | <b>23</b> |
| <b>4.4 The Institutional Silence Problem .....</b>                    | <b>24</b> |
| <b>5. Conclusions and Recommendations.....</b>                        | <b>26</b> |

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>5.1 Conclusions.....</b>                        | <b>26</b> |
| <b>5.2 Recommendations.....</b>                    | <b>26</b> |
| <b>5.2.1 For Government and Regulators .....</b>   | <b>26</b> |
| <b>5.2.2 For Banks and Fintech Platforms .....</b> | <b>27</b> |
| <b>5.2.3 For Government Agencies .....</b>         | <b>28</b> |
| <b>5.2.4 Cross-Sector Recommendations .....</b>    | <b>29</b> |
| <b>References.....</b>                             | <b>31</b> |

## ABSTRACT

Between March and April 2026, a hacker known as ByteToBreach compromised Nigeria's most valuable data stores, the Federal Government's third-party Treasury Single Account platform (Remita), one of Nigeria's largest banks (Sterling Bank), government agency holding company (Corporate Affairs Commission), and national power distributor (Ikeja Electric). Attacks succeeded by targeting a single known vulnerability, which had been undiscovered for 92 days at Remita and three months at Sterling Bank. Risk Across Africa applies tech industry-standard forensic analysis methods to unpack Nigeria's worst-case breach scenario, sector-by-sector, to understand how such incidents can be prevented in the future.

This paper assesses how Nigerian institutions across banking, government, and critical infrastructure sectors can implement resilient IT infrastructure and risk management processes to minimise cyber risk. Applying a comparative case study methodology and guided by analytic framework from the NIST Cybersecurity Framework (2018), Information Technology Resilience Theory, and the Technology Acceptance Model, this paper unpacks vulnerabilities exposed by Nigeria's largest-yet data breach(es), identifies institutional failings that enabled successful exploits across three sectors, and contextualises attacks against a rapidly developing institutional cybersecurity landscape, including the Cybercrimes (Amendment) Act 2024, CBN Risk-Based Cybersecurity Framework (2024), and Critical National Information Infrastructure Order (2024). Findings show that Nigeria's resilience gap is primarily an organisational problem, not technical. Nigerian organisations lack resilient IT infrastructure because they lack the competent, empowered governance necessary to build and maintain it.

# **1. Introduction**

## **1.1 Background**

The digital economy of Nigeria has transformed dramatically in recent years. The rapid adoption of digital technologies across government services, mobile and online banking, fintech innovation, and smart-city infrastructure has established Nigeria as Africa's most digitally engaged market. Utilities like Remita and government databases like the Corporate Affairs Commission, which processes registrations for every business, and bank account in Nigeria, are now critical economic infrastructure. But adoption has not been matched by investment in cybersecurity governance, whether in technical infrastructure or the professionals who maintain it.

Cybersecurity regulations have not kept pace with technological advancement, either. Despite initiatives such as the NDPC's Cybersecurity Policy Framework (2021), the Nigeria Data Protection Regulation (NDPR) enacted by NITDA, the Nigeria Communications Commission's Cybersecurity Regulations for Telecoms Operators 2020, and added DNSF protections implemented since 2020, data controllers face significant grey areas and limited external guidance when developing their cybersecurity strategy.

In practice, this governance gap has been catastrophic. Nigeria is a top-three target in Africa for cyberattacks, with attacks against Nigerian institutions occurring at a rate of just under 5,000 times per week (Check Point Research, 2025). Between 2017 and 2023 alone, cybercrime cost Nigeria approximately N1.1 trillion (\$805 million USD). Financial sector losses from fraud increased 115% between 2024 and 2025 (UNODC, 2025). These incidents are symptomatic not just of threat actor activity but deficiencies in cybersecurity hygiene across Nigeria's banking, government, and critical infrastructure sectors.

The existing cybersecurity literature on Nigeria focuses on technical solutions to combat evolving cybersecurity threats. While fine-grained analyses of technical cybersecurity risk are valuable, institutional weakness presents a broader and less explored threat to Nigeria's ability to safeguard digital assets.

## **1.2 The ByteToBreach Incident: A Cross-Sector Case Study**

On March 18th, 2026, ByteToBreach, a cybercriminal hacker known for breaches in Singapore and Vietnam, announced via press statement that they had accessed and were selling records for every Nigerian adult alive today. The hacker responsible for one of Nigeria's most significant cybersecurity incidents to date was selling all data on the dark web for \$30 million USD.

By March 31st, further details about the breach had been published by those who broke the news: Sterling Bank first suffered a compromise of a server running three outdated software packages with one critical maximum severity vulnerability publicly disclosed but unpatched 92 days prior to the breach.

Technical research led by one of RiskAcrossAfrica's Forensic Analysts uncovered how hacking multiple institutions may have been possible by one threat actor in such rapid succession:

**Sterling Bank:** CVE-2025-55182 allowed initial access via remote code execution on an internet-facing server. ByteToBreach then installed Sliver, a notorious Linux-based command-and-control framework used by hackers to remain undetected in target systems. The hacker remained in Sterling Bank's internal network undetected for nine days, during which time they located plaintext login credentials for Remita's servers stored inside a code repository and used them to gain initial access to Nigeria's Federal Government TSA Platform.

**Corporate Affairs Commission:** Via a vulnerable JWT generation endpoint exposed on the internet. (BizWatch Nigeria, 2026)

**Ikeja Electric:** Through corruption of virtualised infrastructure allowing a threat actor to map internal systems. Unsecured systems connected to external-facing platforms at Ikeja Electric led to doubling-down of efforts by ByteToBreach who returned on April 15th to later claim responsibility for this breach as well.

### 1.3 Timeline of Events

| Name of Institution | Date of Disclosure by ByteToBreach | Stage of Incident | Known to Institution Before Breach (Yes/No) | System Details                                                                                                                           |
|---------------------|------------------------------------|-------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Remita              | March 18                           | Initial Access    | Yes                                         | Production credentials found on Sterling Bank's systems stored in plaintext on a code repository.                                        |
| Sterling Bank       | March 31                           | Persistence       | No                                          | Pilot server running vulnerable versions of three software packages; CVE-2025-55182 was known prior to the attack (CVSS score: 10.0).    |
| CAC                 | April 15                           | Impact            | No                                          | Several vulnerabilities found on port 80, including a vulnerable JWT generation endpoint (/createJWT) leading to CAC API key disclosure. |
| Ikeja Electric      | April 29                           | Remediation       | No                                          | Compromised virtualized infrastructure, including mappings of Siemens-connected metering platforms.                                      |

*Note: Stages adapted from the NIST Cybersecurity Framework (CSF) (2021).*

What made this campaign different from earlier Nigerian cyber incidents was not only its scale, but the response of the institutions that followed. Sterling Bank said nothing to its customers. Remita said nothing to its customers. CAC issued a public statement so understated it bore little resemblance to what had actually occurred. None of the affected organisations apologised (TechNext24, 2026). All of this transpired in clear violation of the Nigeria Data Protection Act 2023. ByteToBreach later confirmed that Sterling Bank had entered into ransom negotiations at a proposed settlement of EUR 250,000 but repeatedly delayed, whereupon the actor published all exfiltrated data on criminal forums (Security Intelligence Substack, 2026).

| Organisation         | Sector                   | Entry Point                                                               | Data Compromised                                                                                                | Institutional Response                                                                                    |
|----------------------|--------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Sterling Bank</b> | Banking                  | CVE-2025-55182 on unpatched pilot server (known 3 months; CVSS 10.0)      | ~900K customer records: BVNs, KYC documents, transaction histories, staff records including MD personal details | No public disclosure; entered covert ransom negotiation at EUR 250,000; data published on criminal forums |
| <b>Remita</b>        | Fintech / Government TSA | Plaintext production credentials found in Sterling Bank's code repository | ~3TB of data; government payment records, TSA transaction data, civil service salary data                       | No public disclosure; no customer notification; no independent incident response                          |
| <b>CAC</b>           | Government Agency        | Vulnerable JWT generation endpoint exploited                              | Company registration records; director data; corporate documents; signatures                                    | Understated public statement issued; NITDA investigation activated; affected individuals not notified     |

|                       |                         |                                                                            |                                                                                                  |                                                                                                             |
|-----------------------|-------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Ikeja Electric</b> | Critical Infrastructure | Ransomware; virtualised infrastructure; Siemens-connected metering systems | Employee passwords; metering platform data; internal directory structure; supplier system access | Spokesperson initially unaware of breach; no formal public disclosure; no regulatory notification confirmed |
|-----------------------|-------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|

Table 1: Summary of ByteToBreach Campaign Incidents, Nigeria (March–April 2026)

## 1.4 Problem Statement

The technical problem is foundational: Nigeria’s institutions are not upgrading critical IT systems or patching disclosed vulnerabilities with adequate urgency.

Second, the governance problem. Almost nobody was held accountable: neither regulators nor the affected institutions believe existing best practice standards such as the Nigeria Data Protection Regulation are worth complying with, or demand genuine consequences for failure.

Third, the structural problem: Private banks are entrusted with access to Nigeria’s government payment infrastructure without meeting the security standards required of federal agencies.

When one fails, we all suffer.

## 1.5 Research Question

**Research Question:** *How can Nigerian organisations across banking, government, and critical infrastructure sectors build resilient IT infrastructure to prevent, detect, and respond to cyberattacks?*

## 1.6 Objectives

The following objectives will guide the investigation to answer this question:

- To conduct root cause analysis of IT infrastructure weaknesses across banking, government, and critical infrastructure sectors exposed by the ByteToBreach incident.
- Compare existing cybersecurity frameworks (e.g. NIST CSF vs. CBN Cybersecurity Framework (2024)) for identified gaps most applicable to Nigeria's cybersecurity context
- Provide actionable recommendations for banks, fintech and government payment platforms, government agencies, and critical infrastructure stakeholders to implement resilient IT infrastructure controls.

## 1.7 Scope, Significance, and Structure

**Sector Scope:** Banking, Government, Critical Infrastructure

**Organisations:** Sterling Bank, Remita, Corporate Affairs Commission (CAC), Ikeja Electric

**Regions:** Nigeria

**Period:** 2024-2026, with case study incidents pulled from March-April 2026 ByteToBreach campaign

Nigeria ranked 5th worst globally for cybercrime on the Cybercrime Index in 2024, with attacks against Nigeria's financial sector increasing 115% YoY between 2024 and 2025. Findings should hold general applicability for Nigerian institutions across all sectors.

### Chapter Structure:

Chapter 2 will assess relevant literature on IT resilience and cybersecurity frameworks developed internationally and within Nigeria.

Chapter 3 outlines RiskAcrossAfrica's approach to investigation.

Chapter 4 presents our findings.

Chapter 5 will outline conclusions and recommendations.

## **2. Literature and Industry Review**

### **2.1 Global Trends in IT Infrastructure Resilience**

Definitions of IT resilience have matured over the last twenty years. In its simplest form, resilience was once treated synonymously with redundancy: the capacity of backup systems to take over upon failure of primary resources; disaster recovery plans put into action. Westerman and Hunter's (2007) framing of IT resilience is more elegant, defining resilience as "the ability to anticipate, absorb, and adapt to events... as well as the capacity to rapidly recover from disruptions while maintaining continuous business operations" (p. 68). This definition presupposes failure and seeks to build a system that is less brittle and unlikely to fail but can cope with failure when it inevitably occurs. It is this principle of elasticity that informs IT resilience frameworks like NIST CSF.

Today's cyber threat landscape has evolved globally in ways that render traditional perimeter-centric security approaches ineffective. Access to ransomware-as-a-service tools has democratized technical expertise; anyone can pay for an attack tool. The International Monetary Fund (2024) estimated the global cost of cyberattacks on the financial sector at \$12 billion over two decades. In the decade after 2020, twice as much money was lost to cyberattacks than in the previous fourteen years. Cloud adoption has expanded the attack surface by introducing vulnerabilities like cloud-storage bucket misconfigurations, API vulnerabilities, and third-party insecure integrations that perimeter defence models cannot detect or remediate. The breach of Remita is a direct consequence of this pattern: approximately three terabytes of Remita customer data was posted to an unsecured Amazon S3 cloud storage bucket (The Guardian Nigeria, 2026).

The limitations of perimeter security models have guided institutional interest in zero trust architecture (ZTA), published by NIST as Special Publication 800-207 in (2020). Zero trust assumes that implicit trust is no longer viable for any user, device, or system attempting to

access resources on your network, including devices already on the network and requires continuous verification of identity and least-privilege access. Every access request is compelled to prove its identity and can only be granted least privilege access for the task at hand. Research evaluating early ZTA adoption within Nigeria’s banking sector revealed a shortfall between formal policy adoption and holistic implementation; gaps in training, process, and cross-functional communication were identified as leading barriers to adoption (Science Publishing Group, 20 25). For organisations who integrate with third parties regularly, like banks integrated with FinTech providers across API interfaces, zero trust is a framework designed to secure against lateral movement – exactly how ByteToBreach moved from Sterling Bank to Remita.

## **2.2 The Nigerian Regulatory and Institutional Landscape**

Nigeria has expended considerable legislative effort towards cybersecurity and privacy regulation. Conceptually, Nigeria builds the requisite legislative scaffolding around key risk vectors. The Cybercrime (Prohibition, Prevention, Etc.) Act was Nigeria’s first omnibus cybersecurity law. Amended in February 2024, organisations of any kind who suffer a cyberattack are required by Section 86 (3) to “with shortest possible delay and not more than 72 hours after becoming aware of the breach...” notify a national incident response centre known as the National Computer Emergency Response Team or ngCERT. Section 83(1) of the same Act mandates that owners of electronic communication facilities allocate resources to cybersecurity measures and incident response (Digital Policy Alert, 2025).

Similarly, Nigeria’s Data Protection Act (NDPA) 2023 placed comprehensive obligations on organisations who own or process data on behalf of consumers. Articles 4(9) and 5(3)(e) both mandate technical security controls be implemented and maintained ‘at all times.’ Following a breach, NDPA section 46 mandates the Nigeria Data Protection Commission (NDPC) and

affected individuals must be notified if the breach ‘is likely to result in a risk to the rights and freedoms’ of those customers. Following its establishment in 2023, the NDPC has displayed a willingness to publicly reprimand and fine erring organisations: N555.8 million were collectively fined by NDPC in 2024 (The Guardian Nigeria, 2025).

The Central Bank of Nigeria addressed the financial sector directly with its Risk Based Cybersecurity Framework Guideline for Deposit Money Banks and Payment Service Banks 2024 mandate. Taking effect on 1 July 2024, the framework mandates governance structures, risk identification and reporting procedures, third-party risk management programs, vulnerability management programs, and cybersecurity controls maturity assessment (CBN, 2024). Compliance with this guideline would have prevented ByteToBreach’s access to Remita through Sterling Bank. API integrations, specifically, are called out by the CBN as introducing risk to ‘poorly managed third party connections’ enabling ‘fraud, API compromise, account takeovers and data privacy violations’ (CBN, 2024). Sterling Bank violated this guideline by failing to patch CVE-2025-55 182. Sterling Bank’s further failure to securely store credentials shared with Remita is a direct violation of section 4(9) NDPA 2023.

On an infrastructure level, the Federal Government published the Designation and Protection of Critical National Information Infrastructure Order (Mondaq, 2024) in June 2024, establishing energy, financial services, healthcare, and education among sectors deemed critical information infrastructure. Subsequently, Lagos State issued a cybersecurity guideline to institutionalise the NIST CSF, ISO/IEC 27001, and other standards as baseline practices following ByteToBreach. Recommendations published in the Cybersecurity Guidelines for Business and Public Institutions (BizWatch Nigeria, 2026) released in April 2026 cite full compliance with NIST CSF 2.0 as a requirement.

Despite these regulatory controls, ByteToBreach exposed gaps between policy and deployed technical controls. In findings shared with BizWatch Nigeria following an independent investigation, cybersecurity consultant David Odes attributed technical failures and organisational blindspots across each of the institutions compromised: “[Responsibility] is on firms and their failure to actually follow through and implement technical controls to address those risks. Because clearly regulations require it but these companies are also failing” (BizWatch Nigeria, 2026). This local affirmation is supported by testimony from the NITDA Director-General at GITEX Africa 2026 that 95% of Nigeria’s cybersecurity incidents are caused by human error.

## **2.3 Sector-Specific Literature**

### **2.3.1 Banking Sector**

Targeting likely reflects their position as frequent customers of banks. Check Point Research (2025) reported cyberattacks targeting Nigerian banks at a monthly average of approximately 18,872, about 630 attacks per day. The Nigeria Inter-Bank Settlement System Plc. ((NIBSS, 2024) recorded fraud notifications from member banks at over N53.4 billion in losses for 2024; one Tier-1 bank lost N10 billion in a single attack. Okonkwo et al. (2025) concluded phishing, social engineering, and unpatched vulnerabilities are prevailing attack vectors in Nigerian cybersecurity incidents after secondary analysis of CBN breach reports and internal bank data. ByteToBreach exploited a known maximum severity vulnerability on a server accessible from the internet. Failure to patch persisted for three months.

### **2.3.2 Government Sector**

Government agencies are victims of unique cybersecurity challenges endemic to systemic underfunding, reliance on legacy technology, and organisational cultures unwilling to publicly

own breaches until forced. Corporate Affairs Commission attracted significant attention following the breach because the stolen data included registrations for every business in Nigeria. Beyond any contractual or citizen claim to privacy, the volume and types of personal data accessed exposes victims to concrete secondary fraud risks: Directors' identity theft, corporate documents forgery, and fraud with compromised signatures/registration photocopies now available on criminal markets (BizWatch Nigeria, 2026). Establishing and continuously maintaining access controls on privileged endpoints like `jwt.cac.ng/register/token` is a basic control likely contained in annual cybersecurity audits if they were happening. If the breached endpoint was not covered by pre-existing controls, it was because they did not exist. Either scenario confirms governance failures at CAC.

### 3.3 Critical Infrastructure

Ikeja Electric could have qualified for inclusion under Remita or Sterling Bank had data been the objective. Ransomware attacks on critical infrastructure also pose threat to national security because they compromise the delivery of essential services. Although physical safety was not digitally threatened in this instance, theft of operational environment configurations poses supply chain risks. Nigeria does not currently have utility or energy-specific cybersecurity standards mandating basic controls. Cybersecurity essentials like network segmentation were also absent. Once converged, IT and OT share vulnerability datasets. Failure to separate IT networks from OT networks physically and logically exposed elements of Ikeja Electric's IT infrastructure to direct compromise.

## 2.4 Theoretical Frameworks

This paper applies four complementary theoretical frameworks:

- **NIST Cybersecurity Framework (CSF):** The NIST CSF offers a vocabulary and organising framework for cybersecurity controls across five functions: **Identify**,

**Protect, Detect, Respond, Recover** (NIST, 2018). Guiding this paper's cross-case comparison, NIST CSF allows for systematic comparison across incident lifecycle stages. The 2026 Lagos Cybersecurity Guidelines explicitly incorporate NIST CSF as the reference framework for incident response planning (Mondaq, 2026).

- **IT Resilience Theory (Westerman & Hunter, 2007):** Westerman and Hunter's work on reframing cybersecurity from rigid defense to dynamic resilience informs this paper's emphasis on weaknesses in detection, response, and recovery. Building resilience requires competencies beyond prevention.
- **Technology Acceptance Model (Davis, 1989):** Organisations are not adopting cybersecurity controls known to support resilience. Why? The TAM offers a lens for examining human behaviour around technology adoption. The central variables of perceived usefulness and perceived ease of use can elucidate common sense cybersecurity practices that resist institutional adoption.
- **Shared Responsibility Model:** Connections between banks and FinTech providers are relationships of shared security responsibilities. Attackers moved between four organisations across three sectors. Assessing how responsibilities were divided, communicated, and enforced is guided by the Shared Responsibility Model.

### **3. Investigative Approach**

#### **3.1 Research Design**

This paper utilises qualitative methods to investigate a topic where little independent research has been published. Case study research is appropriate because the event of interest is bounded (ByteToBreach), occurred in the real-world at-scale, and can be studied in-depth across multiple organisations. Comparative case studies enable both within-case (organic failure patterns within an institution) and cross-case (similarities and differences across institutions) analysis (Yin, 2018). The unit of analysis is Nigerian organisations' IT infrastructure resilience posture, benchmarked against industry standard NIST CSF. The case boundary is defined by campaign start 18 March 2026 and campaign end 30 April 2026.

#### **3.2 Data Sources**

- Investigative and threat intelligence reports: The Web Security Lab investigation into the Sterling Bank–Remita breach chain (Security Intelligence Substack, April 2026); the ByteToBreach interview conducted by Security Intelligence (April 2026); analysis by cybersecurity consultant David Odes as reported by Techloy (April 2026); and the Cyfirma Nigeria Cyber Threat Assessment (September 2025).
- Regulatory and institutional statements: CAC official breach notification (April 2026); NDPC Notice of Investigation (April 2026); CBN Risk-Based Cybersecurity Framework and Guidelines (May 2024); Critical National Information Infrastructure Order (June 2024); Lagos State 2026 Cybersecurity Guidelines.
- Secondary academic and industry literature: Peer-reviewed journals including IJRIAS (2025), IJRISS (2025), and the Journal of Electrical and Electronic Engineering (2025); industry reports from Check Point Research, NIBSS, IMF, and UNODC.

### **3.3 Analytical Tools**

- NIST CSF Gap Analysis: Assessment of each of four institutions across each of five NIST cybersecurity framework functions.
- SWOT Analysis: Each sector (banking, government, critical infrastructure) receives a SWOT matrix to help contextualise conclusions within sector-specific cybersecurity landscapes.
- Root Cause Analysis: Drawing on the threat actor's account and independent technical investigations, root causes are classified along four dimensions: technical, organisational, regulatory, and structural.

### **3.4 Ethical Considerations and Limitations**

- Public data: This study is limited to publicly available information. Statements made by the ByteToBreach threat actor are not independently verified facts and are treated as claims requiring additional technical forensics to validate. Affected individuals whose data was posted on hacking forums are not named.
- Availability of information: The full internal incident response and investigation details of each affected organisation are not public. Public reports and statements have been used to fill gaps. This limits the depth of analysis possible for the Respond and Recover categories.

## 4. Findings and Analysis

### 4.1 NIST CSF Gap Analysis: Cross-Case Findings

Table 3 maps the documented failures of each organisation against the five NIST Cybersecurity Framework functions. The analysis draws directly on investigation reports, the Security Intelligence interview with the threat actor, regulatory statements, and independent cybersecurity analyses.

| NIST Function                                       | Sterling Bank                                                                                                                                                   | Remita                                                                                                                             | CAC                                                                                            | Ikeja Electric                                                                                                             |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>IDENTIFY (Asset Mgmt, Risk Assessment)</b>       | Pilot server in asset inventory but treated as low-risk; CVE-2025-55182 on record for 3 months with no escalation; third-party credential exposure not assessed | No independent risk assessment of Sterling Bank relationship; TSA credentials not classified as critical assets requiring vaulting | JWT endpoint not identified as high-risk; incomplete mapping of internet-facing attack surface | OT/IT convergence not mapped; Siemens-connected metering systems not assessed as IT assets subject to cybersecurity policy |
| <b>PROTECT (Access Control, Patching, Training)</b> | Maximum-severity CVE unpatched for 3 months; no MFA enforcement; Remita credentials stored in plaintext in code repository                                      | Production credentials in plaintext in code repository; no independent authentication layer for TSA integration                    | Vulnerable JWT endpoint unpatched; insufficient access controls on registration system         | No network segmentation between IT and OT; no access controls on metering platform; virtualised infrastructure unsegmented |
| <b>DETECT (Monitoring, Anomaly Detection)</b>       | Attacker persisted 9 days undetected; breach discovered                                                                                                         | No independent detection capability; breach discovered                                                                             | Breach detected post-exploitation; no real-time intrusion                                      | No IDS/IPS on metering or OT systems; ransomware deployed without                                                          |

|                                                          |                                                                                                                                       |                                                                                                       |                                                                                                                   |                                                                                                                |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                                          | externally via criminal forum post, not internal monitoring                                                                           | through Sterling Bank's criminal forum publication                                                    | detection on registration system                                                                                  | triggering automated response                                                                                  |
| <b>RESPOND<br/>(Incident Response, Notification)</b>     | No customer notification; ransom negotiation conducted covertly at EUR 250,000; NDPA 2023 and 72-hr ngCERT requirements not fulfilled | No customer notification; no independent incident response; fully reliant on Sterling Bank's response | Public statement issued but materially understated incident; NITDA activated; customer notification not confirmed | Spokesperson initially unaware of breach; no formal public disclosure; no evidence of stakeholder notification |
| <b>RECOVER<br/>(Recovery Plan, Post-Incident Review)</b> | No recovery plan disclosed; data published on criminal forums; ransom outcome unknown                                                 | No recovery plan disclosed; government financial operations impact not publicly assessed              | JWT endpoint suspended; NITDA engagement ongoing; full recovery timeline unknown                                  | No formal recovery disclosure; ongoing as of late April 2026; no post-incident review published                |

*Table 3: NIST CSF Gap Analysis, ByteToBreach Affected Organisations (March–April 2026)*

The gap analysis reveals a consistent pattern across all four organisations: failure is concentrated in the Protect and Respond functions. All four had some form of identification framework, asset inventories, regulatory obligations, but failed to translate identified risks into protective action. The Remita case is instructive: the company's failure was not its own Protect function but its complete absence from the Identify function applied to its third-party relationships. Remita's risk register evidently did not include 'plaintext production credentials sitting in Sterling Bank's code repository' as a threat scenario, because Remita had never audited Sterling Bank's handling of those credentials.

The Detect function failures are equally systemic. ByteToBreach persisted in Sterling Bank's systems for nine days before discovery, not through Sterling Bank's own monitoring, but through a post on a criminal forum. Basic SIEM systems configured to flag unusual outbound data transfers or new administrative account creations would have identified the intrusion within hours. The absence of such capabilities in institutions subject to the CBN's 2024 Cybersecurity Framework, which explicitly requires continuous monitoring, suggests compliance is being treated as a documentation exercise rather than an operational standard.

## **4.2 Root Cause Analysis**

### **4.2.1 Technical Root Causes**

CVE-2025-55182 was a known vulnerability. It was known to Sterling Bank; they had it on record. It was maximum severity, CVSS score 10.0, disclosed publicly by the vendor, and had been for months. The vulnerability was accessible from the public internet. Anyone who knew it was there could reach it and send a specially crafted request to make Sterling Bank's server do anything they wanted (TechNext24, 2026; Security Intelligence Substack, 2026). The threat actor responsible for this campaign.

Failures in authentication and credential management are equally consequential. Remita's production credentials were openly accessible to Sterling Bank, and stored in plaintext on Sterling Bank's servers. Basic controls like multi-factor authentication (MFA) and password hashing are not adopted universally. This fact, more than any guideline or security architecture framework, exposes why resilience fails even when institutions understand its components: IT security leaders know these controls work, but the organisations they work for do not make them standard practice.

#### **4.2.2 Organisational Root Causes**

Failure to patch is never just a technical problem. Every Nigerian organisation with functional IT security teams knows what to do because they know what not to do. The three-month gap between CVE disclosure and exploitation is not merely a technical failure, it is an organisational process failure. A functioning vulnerability management process would have ingested the CVE advisory, assessed its severity (CVSS 10.0), identified affected systems, and escalated remediation within days. That this did not happen indicates the process either did not exist, was not followed, or lacked authority to compel action. Why was CVE-2025-55182 not patched?

The institutional silence that followed all four breaches reveals a governance culture in which reputation management overrides legal obligation. The humans whose data was stolen, bank customers whose BVN numbers and identity documents are on criminal forums; government employees whose salary payment infrastructure was compromised; company directors whose signatures now circulate on dark web markets, were not told. ByteToBreach confirmed the goal was exfiltration and a one-time sale, after failure of extortion (BizWatch Nigeria, 2026). The data was always going to be sold. The silence of institutions only ensured that customers could not take protective action.

#### **4.2.3 Regulatory Root Causes**

Rules exist. The requirements outlined by Nigeria's Central Bank were violated by Nigeria's Central Bank. Failure to detect and then follow responsible disclosure procedures not only violates NCB's 2024 Risk-Based Cybersecurity Framework but basic forensic hygiene. As well as the Cybercrimes (Amendment) Act 2024. Rules have consequences. Compliance must be measurable, reportable, and enforced publicly. Until Nigeria establishes consistently applied

consequences for non-compliance with its cybersecurity regulations, junior officers will treat disclosed vulnerabilities as optional risk mitigations, not mandatory technical requirements.

#### 4.2.4 Structural Root Causes

Shared technology ecosystems heighten risks when third-party integrations lack formal security obligations. Remita operates Nigeria's Treasury Single Account (TSA), a payment portal handling monetary transactions on behalf of the Nigerian government. Connection to Remita alone should have elevated Sterling Bank's internal handling of TSA authentication credentials to a national security risk. Whether by CBN mandate or private contract agreement, third-party posture should have been scoped, assessed, and monitored. Instead, Remita never audited Sterling Bank cybersecurity practices. Sterling Bank never considered Remita's security requirements when storing credentials in plaintext. Neither party had formally documented what the other should have done about cybersecurity.

### 4.3 Sector SWOT Analysis

| Sector                                      | Strengths                                                                                                                                     | Weaknesses                                                                                                                                                                    | Opportunities                                                                                                                                               | Threats                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BANKING<br/>(Sterling Bank / Remita)</b> | CBN Cybersecurity Framework 2024; NIBSS fraud monitoring; growing AI fraud detection investment; some banks have established SOC capabilities | CBN compliance as documentation not operation; third-party risk management absent; credential management failures; insider fraud N19bn (2024); culture of silence post-breach | Zero-trust architecture pathway; GITEX Africa partnerships; ISO 27001 adoption for cross-border requirements; AI-powered SOC tools becoming cost-accessible | RaaS attacks requiring no specialist skill; dark web credential markets; interconnection with government infrastructure amplifies single-point-of-failure risk; NDPA enforcement risk |

|                                                     |                                                                                                                                                       |                                                                                                                                                                               |                                                                                                                                                          |                                                                                                                                                                             |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GOVERNMENT<br/>(CAC / NITDA)</b>                 | NITDA coordination capacity; NDPA 2023 enforcement framework; NDPC demonstrated enforcement (N555.8m fines 2024); Lagos 2026 Cybersecurity Guidelines | Chronic underfunding; legacy system dependency; no breach disclosure culture; human error rate 95%; leadership awareness gap; JWT endpoint left vulnerable                    | National CSIRT development; CNII Order 2024 mandate; GITEX Africa partnerships; UNODC technical assistance; AU Convention on Cybersecurity (signed 2024) | Politically motivated attacks on electoral infrastructure; data sovereignty risks; election data integrity (INEC server); forged corporate documents from CAC breach        |
| <b>CRITICAL INFRASTRUCTURE<br/>(Ikeja Electric)</b> | Growing smart grid investment; Siemens partnership brings technical capacity; CNII designation (2024) creates political mandate for protection        | No OT-specific cybersecurity standards; IT/OT convergence without security controls; no IDS on operational systems; executive awareness gap; no tested incident response plan | Sector-specific CERT for energy; NERC CIP or NIS2-equivalent standards; vendor security requirements; World Bank infrastructure security programmes      | Ransomware on utility systems can disrupt millions; supply chain attacks through vendor systems; physical safety consequences of OT compromise; limited recovery capability |

*Table 4: Cross-Sector SWOT Analysis, Nigeria Cybersecurity, 2026*

#### 4.4 The Institutional Silence Problem

The most meaningful data point to emerge from this cross-case comparison may be what didn't happen. After exploiting vulnerabilities and stealing data from these four organisations, after dedicating weeks to intrusion maintenance before they were caught by a stranger on hacking forums, after selling sensitive government data and potentially every Nigerian bank customer's

database to the highest bidder: these institutions chose not to tell the customers whose data was stolen.

ByteToBreach confirmed he contacted all three organisations and offered to delete stolen data in exchange for money: Sterling Bank heExtorted (Bitcoin worth, at the time, EUR 250, 000), NGX, CAC ( BusinessDay, 2026). Customers were not told NG.Exit the window of opportunity to change passwords, monitor accounts, cancel BVNs, or take precautions with direct knowledge of breach likelihood. They were told nothing.

This silence has three dimensions. First, it is illegal: NDPA 2023 requires notification without undue delay upon becoming aware of a breach likely to result in risk to data subjects. The Cybercrimes (Amendment) Act 2024 requires ngCERT notification within 72 hours. None of the affected organisations met these obligations. Second, it is strategically counterproductive: customers who are not notified cannot change passwords, monitor accounts, freeze credit, or report suspicious documents. Third, it reflects a governance culture in which data subjects are treated as reputational liabilities rather than stakeholders whose interests must be protected. The ByteToBreach campaign, the most publicly documented Nigerian cyber incident to date, is an opportunity for regulatory enforcement to demonstrate that this culture carries consequences.

## **5. Conclusions and Recommendations**

### **5.1 Conclusions**

To reiterate: what started this research was a question about how Nigerian institutions can protect their IT infrastructure. The answer, inevitably, is wherever ByteToBreach was not: not in Sterling Bank's changelogs, not in Remita's dev team's password managers, not in CCTV footage of an unknown person infiltrating Ikeja Electric's facilities. Every shortcoming of Nigeria's cybersecurity posture demonstrates where ByteToBreach succeeded. Table 4 distills cross-case findings into accessible observations.

Fundamentally, Nigeria doesn't lack cybersecurity know-how, we lack adherence to known cybersecurity best practice. Encryption is standard practice. MFA is routine. Organisations know software upgrades are necessary until they don't. Patching Server vulnerability on web was not done because someone at Sterling Bank knows better.

It doesn't matter if Nigeria drafts perfect cybersecurity regulations ten times. Until compliance failures consistently result in consequences that threaten CEOs' livelihoods, Nigeria will not see enduring improvements to cybersecurity resilience.

NDPC can fine organisations all they want, but IT resilience is cross-sector. Regulation should be too.

### **5.2 Recommendations**

#### **5.2.1 For Government and Regulators**

Regional specifics and landscape nuance from local sources that references established literature or surrounding research from this topic area.

Regional policymakers and legislators should advocate holistic frameworks that incentivise cross-sector collaboration around cybersecurity in practice, not theory

Cross-sector standards should establish open communication standards between parties where data or security responsibilities are shared. Remediation fails when human beings fail to communicate.

Elected leaders should have represented customers affected by these breaches. Wherever democratic action is possible, use your voice. If a concession stand thief stole your \$20, would you allow them to keep it? Phone manufacturers ought to investigate engineering empathy skills into SIEM alerting workflows. Frankly, so should someone.

- Establish a National Cyber Incident Response Team (NCIRT) with cross-sector coordination authority and a mandate to conduct post-incident reviews of significant national breaches, publishing findings as public-interest cybersecurity guidance.
- Mandate OT/ICS-specific cybersecurity standards for all CNII operators in the energy sector, modelled on NERC CIP (North America) or the EU's NIS2 Directive, with compliance timelines and enforcement mechanisms.
- Require all government agencies holding citizen data to undergo independent cybersecurity audits every 12 months, with classified vulnerabilities remediated within defined timelines under regulatory oversight.
- Establish a Shared Threat Intelligence Platform (STIP) linking CBN, NITDA, NCC, ngCERT, and critical infrastructure operators, enabling real-time sharing of attack indicators, CVE exploitation alerts, and sector-specific threat assessments.

### **5.2.2 For Banks and Fintech Platforms**

Sterling Bank's breach demonstrates that basic security hygiene failures remain the primary attack vector for Nigerian financial institutions, despite the sophistication of regulatory frameworks governing them.

- Implement a formal vulnerability management programme with mandatory remediation timelines graduated by CVSS severity. CVSS 9.0+ vulnerabilities on internet-facing systems must be patched within 72 hours of CVE publication, not three months.
- Adopt zero-trust architecture principles for all third-party integrations. No implicit trust for any system, device, or user; continuous verification of identity and least-privilege access for all FinTech and government API connections.
- Implement credential vaulting for all production credentials. No production credentials to be stored in code repositories, configuration files, or any unencrypted location. Mandatory adoption of secrets management tools (HashiCorp Vault, AWS Secrets Manager, or equivalent).
- Establish a 24/7 Security Operations Centre (SOC) with SIEM capabilities configured to detect unusual outbound data transfers, new administrative account creations, and lateral movement. The nine-day undetected presence of ByteToBreach in Sterling Bank's systems is incompatible with any functional monitoring capability.
- Conduct contractually mandated third-party security assessments for all platforms with access to government infrastructure, with findings shared with CBN as part of risk-based cybersecurity reporting.

### **5.2.3 For Government Agencies**

Government agencies present a distinct risk profile characterised by high-value data holdings, legacy system dependency, and structural underfunding. Recommendations must be achievable within resource constraints while addressing the most critical vulnerabilities.

- Implement a mandatory security-by-design standard for all new government digital infrastructure, requiring independent security review before deployment. JWT endpoints, API integrations, and authentication systems must be assessed against OWASP Top 10 standards before going live.

- Launch a mandatory cybersecurity awareness programme for all government agency executives and board-level officials. The Ikeja Electric spokesperson's unawareness of the breach is symptomatic of a leadership gap that technical controls alone cannot address.
- Allocate dedicated, protected cybersecurity budget lines in all agency annual budgets. NITDA should establish minimum cybersecurity investment thresholds as a condition of regulatory compliance for agencies designated as CNII operators.
- Establish incident response plans for all agencies holding citizen data, with mandatory annual tabletop exercises and results reported to NITDA. Plans must include specific notification procedures for NDPC, ngCERT, the public, and affected individuals.

#### **5.2.4 Cross-Sector Recommendations**

The cross-sector nature of the ByteToBreach campaign demands cross-sector solutions that individual institution-level improvements cannot resolve.

- Formalise a Shared Security Standard for all public-private infrastructure integrations, beginning with the banking-government payment interface. Any private bank integrated with Remita or the TSA should be required to meet a defined security baseline, including vulnerability management timelines and credential handling standards, as a condition of maintaining integration access.
- Commission a National IT Resilience Audit of all CNII organisations, assessing baseline posture against NIST CSF across all five functions, and informing a five-year national cyber resilience roadmap with measurable targets.
- Institute the ByteToBreach campaign as a mandatory case study in all public sector cybersecurity training, banking sector induction programmes, and postgraduate IT management curricula. Real, documented, consequential failures are the most effective instructional material available.

- Formally adopt and ratify the African Union Convention on Cybersecurity and Personal Data Protection (signed January 2024), activating international cooperation and capacity-building provisions, and aligning domestic frameworks with continental standards.

## References

BizWatch Nigeria. (2026, April 17). Explainer: ByteToBreach hacker sheds light on why

CAC, Sterling Bank and Remita were targeted. BizWatch Nigeria.

<https://bizwatchnigeria.ng/explainer-bytetobreach-hacker-sheds-light-on-why-cac-sterling-bank-and-remita-were-targeted/>

Central Bank of Nigeria. (2024, May 31). Risk-based cybersecurity framework and

guidelines for deposit money banks and payment service banks. CBN.

[https://www.cbn.gov.ng/Out/2024/BSR/CBN%20Risk-Based%20Cybersecurity%20Framework%20for%20DMBs%20and%20PSBs\\_2024.pdf](https://www.cbn.gov.ng/Out/2024/BSR/CBN%20Risk-Based%20Cybersecurity%20Framework%20for%20DMBs%20and%20PSBs_2024.pdf)

Check Point Research. (2025). 2025 cyber security report: Nigeria threat landscape. Check

Point Software Technologies.

Culture Custodian. (2026, April 28). ByteToBreach: The unravelling of Nigeria's digital

infrastructure. The Culture Custodian. <https://culturecustodian.com/bytetobreach-the-unravelling-of-nigerias-digital-infrastructure/>

Cyfirma. (2025, September). Cyber threat assessment on Nigeria. Cyfirma Research.

<https://www.cyfirma.com/research/cyber-threat-assessment-on-nigeria/>

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of

information technology. MIS Quarterly, 13(3), 319-340.

<https://doi.org/10.2307/249008>

Digital Policy Alert. (2025). DPA digital digest: Nigeria [2025 edition]. Digital Policy Alert.

<https://digitalpolicyalert.org/digest/dpa-digital-digest-nigeria>

Economy Post. (2026, April 28). Ikeja Electric suffers data breach, targeted in ransomware attack. Economy Post Nigeria. <https://economypost.ng/technology/ikeja-electric-suffers-data-breach-targeted-in-ransomware-attack/>

Federal Government of Nigeria. (2024, June). Designation and protection of critical national information infrastructure order. Office of the National Security Adviser.

Further Africa. (2024, December 4). Safeguarding Nigeria's critical national information infrastructure. Further Africa. <https://furtherafrica.com/2024/12/04/safeguarding-nigerias-critical-national-information-infrastructure/>

The Guardian Nigeria. (2026, April 20). Election data at risk as hackers hold government agencies, banks to ransom. The Guardian Nigeria. <https://guardian.ng/featured/election-data-at-risk-as-hackers-hold-govt-agencies-banks-to-ransom/>

International Journal of Research and Applied Sciences (IJRIAS). (2025). Integrating governance, AI accountability, and systemic resilience: A framework for critical infrastructure protection in Nigeria. RSIS International. <https://rsisinternational.org/journals/ijrias/view/integrating-governance-ai-accountability-and-systemic-resilience-a-framework-for-critical-infrastructure-protection-in-nigeria>

International Journal of Research and Innovation in Social Science (IJRISS). (2025, July 31). Protecting critical infrastructure in Nigeria: A framework for integrated cybersecurity approach. RSIS International. <https://rsisinternational.org/journals/ijriss/articles/protecting-critical-infrastructure-in-nigeria-a-framework-for-integrated-cybersecurity-approach/>

International Monetary Fund. (2024). Global financial stability report: The last mile. IMF Publication Services.

Legit.ng. (2026, April 16). CAC confirms cyberattack, system breach. Legit.ng.

<https://www.legit.ng>

Mondaq. (2024, December). Overview of the CBN risk-based cybersecurity framework and guidelines for deposit money banks and payment service banks. Mondaq Nigeria.

<https://www.mondaq.com/nigeria/security/1518574>

Mondaq. (2026, May). A review of the 2026 cybersecurity guidelines for business and public institutions in Lagos State. Mondaq Nigeria.

<https://www.mondaq.com/nigeria/telecoms-mobile-cable-communications/1786220>

National Institute of Standards and Technology. (2018). Framework for improving critical infrastructure cybersecurity (Version 1.1). U.S. Department of Commerce.

<https://doi.org/10.6028/NIST.CSWP.04162018>

National Institute of Standards and Technology. (2020). Zero trust architecture (NIST Special Publication 800-207). U.S. Department of Commerce.

<https://doi.org/10.6028/NIST.SP.800-207>

Nigeria Inter-Bank Settlement System. (2024). Annual fraud and forgeries report 2024.

NIBSS.

Okonkwo, C. J., & others. (2025). Cybersecurity in Nigerian deposit money banks: Attack vectors and mitigation strategies. International Journal of Science and Research Archive, 15(02), 758-761.

OgunSecurity / OSRS Intelligence. (2026, April 15). Africa's digital infrastructure is under attack and the consequences are irreversible.

<https://www.ogunsecurity.com/post/africa-s-digital-infrastructure-is-under-attack-and-the-consequences-are-irreversible>

Science Publishing Group. (2025, June 23). Evaluating the application of Zero Trust Architecture (ZTA) implementation in Nigeria's banking industry. Journal of Electrical and Electronic Engineering, 13(3).

<https://www.sciencepublishinggroup.com/article/10.11648/j.jeee.20251303.12>

Security Intelligence Substack / Web Security Lab. (2026, April 8). Sterling Bank and Remita: How a global hacker walked through Nigeria's banking sector and took everything. <https://securityintelligence.substack.com/p/sterling-bank-and-remita-how-a-global-f9c>

Security Intelligence Substack / Web Security Lab. (2026, April 16). I spoke with ByteToBreach: The CAC breach, Sterling Bank's EUR 250,000 ransom, and why Nigeria. <https://securityintelligence.substack.com/p/i-spoke-with-bytetobreach-the-cac>

Security Intelligence Substack / Web Security Lab. (2026, April 29). A security advisory for every Nigerian organisation still making itself a soft target. <https://securityintelligence.substack.com>

The Cable. (2025, March 26). Assessing CBN's cybersecurity framework and guidelines. TheCable Nigeria. <https://www.thecable.ng/assessing-cbns-cybersecurity-framework-and-guidelines/>

Techloy. (2026, April 22). Inside the ByteToBreach campaign: How a single vulnerability compromised a nation's financial infrastructure. Techloy. <https://www.techloy.com/inside-the-bytetobreach-campaign-how-a-single-vulnerability-compromised-a-nations-financial-infrastructure/>

TechNext24. (2026, April 17). Sterling Bank, Remita and CAC data breaches: Why did the Nigerian institutions say nothing? TechNext24.

<https://technext24.com/2026/04/17/sterling-bank-remita-cac-data-breach/>

Triumph Times. (2026, April 17). An analysis of the CAC, Remita, and Sterling Bank data breaches: Risks, failures, and lessons. Triumph Times Nigeria.

<https://www.triumphtimes.com/2026/04/17/an-analysis-of-the-cac-remita-and-sterling-bank-data-breaches-risks-failures-and-lessons/>

United Nations Office on Drugs and Crime. (2025). Nigeria cybercrime assessment 2025. UNODC.

Vanguard. (2025, January 1). Nigeria, SA, Algeria top targets for cyber attacks in 2024. Vanguard Newspapers Nigeria.

Westerman, G., & Hunter, R. (2007). IT risk: Turning business threats into competitive advantage. Harvard Business School Press.

Yin, R. K. (2018). Case study research and applications: Design and methods (6th ed.). SAGE Publications.